Customer Churn Analytics

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Objective: Customer Churn Analytics For Banking Industry. A Comprehensive Study on Customer Retention Drivers and Risk Factors

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Executive Summary

Problem Statement:
The banking organization faces a critical customer retention challenge with an overall churn rate of 20.37% (2,037 out of 10,000 customers). This represents significant revenue leakage and elevated customer acquisition costs required to maintain portfolio growth.

Key Findings:
Germany exhibits the highest churn rate at 32.44%, more than double that of France (16.15%) and Spain (16.67%), indicating a critical geographic risk.

The 56-65 age group faces extreme churn at 48.32%, representing the highest risk demographic segment requiring immediate targeted intervention.

Inactive customers churn at 26.85% compared to active customers at 14.27%, demonstrating that engagement status is a strong churn predictor.

Executive Summary

Customers with 3-4 products show anomalously high churn rates (82.71% and 100%), suggesting potential service complexity or satisfaction issues requiring investigation.

Female customers exhibit higher churn (25.07%) than male customers (16.46%), indicating a need for gender-specific retention strategies.

Zero-balance accounts show significantly lower churn (13.82%) compared to accounts with any balance (23-24%), suggesting dormant accounts may not represent true attrition risk.

Business Impact:
The concentration of churn in high-value segments (Germany, older customers, high-balance accounts) indicates significant risk to revenue. Additionally, the extreme churn among customers with multiple products suggests potential service delivery or product fit issues.

Dataset Overview

- The analysis uses a bank customer dataset with 10,000 records.
- Each record represents one retail banking customer with attributes covering demographics, relationship details, product usage, and churn status.
- Key demographic variables include Geography, Gender, and Age.
- Relationship and behavior variables include Tenure with the bank, Number of Products held, HasCrCard (credit card ownership), and IsActiveMember (account activity status).
- Financial variables include Balance, EstimatedSalary, and CreditScore.
- The target variable is Exited (True/False), indicating whether the customer has left the bank, and all analyses focus on understanding how churn varies across these features.

EDA: Exploratory Data Analysis

Demographic profile:

- **The dataset covers 10,000 customers across three geographies, with France having 5,014 customers, Germany 2,509, and Spain 2,477, and a gender split of 4,543 females and 5,457 males.**
- **Age is concentrated in working-age groups, with 4,153 customers aged 18–35 and 5,047 aged 36–55, while only 800 customers are 56+ (536 aged 56–65 and 264 aged 65+).**

	geography character varying (50)	customercount bigint
1	France	5014
2	Germany	2509
3	Spain	2477

	gender character varying (10)	customercount bigint
1	Female	4543
2	Male	5457

	agebucket text	customercount bigint
1	18-35	4153
2	36-55	5047
3	56-65	536
4	65+	264

EDA: Exploratory Data Analysis

Financial profile:

- **Customers have an average balance of about 76k with a median of about 97k, indicating a right-skewed distribution where many customers have lower balances and a smaller group holds very high balances.**

	customerbalance bigint	averagebalance numeric	medianbalance numeric	minbalance numeric	maxbalance numeric	stddevbalance numeric
1	10000	76485.89	97198.54	0.00	250898.09	62397.41

- **Estimated salary averages around 100k with a wide spread (standard deviation ≈ 57.5k), and credit scores cluster around a mean of 651, spread across Poor, Average, Good, and Excellent buckets.**

	customercount bigint	averageestimatedsalary numeric	medianestimatedsalary numeric	minestimatedsalary numeric	maxestimatedsalary numeric	stddevestimatedsalary numeric
1	10000	100090.24	100193.92	11.58	199992.48	57510.49

	customercount bigint	averagecreditscore numeric	mediancreditscore numeric	mincreditscore numeric	maxcreditscore numeric	stddevcreditscore numeric
1	10000	650.53	652.00	350.00	850.00	96.65

EDA: Exploratory Data Analysis

Behavioral profile:

- **Relationship length is well distributed, with 1,448 customers having tenure 0–1 years, 3,046 with 2–4, 3,007 with 5–7, and 2,499 with 8–10 years at the bank.**
- **Product usage and engagement show that most customers hold 1–2 products (9,674 customers), 7,055 have a credit card, 5,151 are active members, and 4,849 are not active, giving a balanced mix for behavior-based churn analysis**

	tenurebucket text	customercount bigint
1	0-1	1448
2	2-4	3046
3	5-7	3007
4	8-10	2499

	numofproducts integer	customercount bigint
1	1	5084
2	2	4590
3	3	266
4	4	60

	hascrcard boolean	customercount bigint
1	false	2945
2	true	7055

	isactivemember boolean	customercount bigint
1	false	4849
2	true	5151

Key Churn Drivers

Demographic drivers:

	geography character varying (50)	churncount bigint	totalcustomers bigint	churnpercentage text
1	France	810	5014	16.15%
2	Germany	814	2509	32.44%
3	Spain	413	2477	16.67%

- **Geography – Germany (32.44%) vs France (16.15%) and Spain (16.67%).**

	gender character varying (10)	churncount bigint	totalcustomers bigint	churnpercentage text
1	Female	1139	4543	25.07%
2	Male	898	5457	16.46%

- **Gender – Females (25.07%) vs Males (16.46%).**

	agebucket text	exitedcount bigint	totalcustomers bigint	exitedpercentage text
1	18-35	347	4153	8.36%
2	36-55	1396	5047	27.66%
3	56-65	259	536	48.32%
4	65+	35	264	13.26%

- **Age group – 56–65 age group shows the highest churn at 48.32%.**

Key Churn Drivers

Financial drivers:

- **Account balance – Zero balance customers churn less (13.82%) than those with positive balances (around 23–24%).**

	balancebucket text	exitedcustomers bigint	totalcustomers bigint	exitedpercentage text
1	0	500	3617	13.82%
2	1-75k	101	424	23.82%
3	75k-150k	1212	4990	24.29%
4	150k+	224	969	23.12%

- **Credit score – Churn stays around 19–23% across Poor, Average, Good, and Excellent segments, indicating a weak relationship.**

	salarybucket text	exitedcustomers bigint	totalcustomers bigint	exitedpercentage text
1	Low income 20 Percentile	424	2110	20.09%
2	Lower-middle income 40 Percen...	217	1102	19.69%
3	Middle income 60 Percentile	191	982	19.45%
4	Upper-middle income 80 Percen...	409	2004	20.41%
5	High income 100 Percentile	796	3802	20.94%

- **Estimated salary – Churn is almost flat (~19–21%) across all income percentile buckets, showing no strong churn variation.**

	creditscorebucket text	exitedcustomers bigint	totalcustomers bigint	exitedpercentage text
1	0-550 (Poor)	368	1621	22.70%
2	551-650 (Average)	689	3316	20.78%
3	651-750 (Good)	667	3465	19.25%
4	751+ (Excellent)	313	1598	19.59%

Key Churn Drivers

Behavioral / Relationship drivers:

- **Tenure – Churn is relatively flat across 0–1, 2–4, 5–7, and 8–10 years (around 19–22%), so tenure alone is not a strong predictor.**

	tenurebucket text	totalexited bigint	totalcustomers bigint	exitedpercentage text
1	0-1	327	1448	22.58%
2	2-4	617	3046	20.26%
3	5-7	582	3007	19.35%
4	8-10	511	2499	20.45%

- **Product portfolio – Customers with 3–4 products have extremely high churn (82.71% and 100%), while those with 2 products churn the least (7.58%).**

	numofproducts integer	exitedcustomers bigint	totalcustomers bigint	exitedpercentage text
1	1	1409	5084	27.71%
2	2	348	4590	7.58%
3	3	220	266	82.71%
4	4	60	60	100.00%

Key Churn Drivers

Behavioral / Relationship drivers:

- **Activity status – Inactive members churn at 26.85% compared to 14.27% for active members, making activity a strong driver.**

	activitystatus text	customercount bigint	exitedcount bigint	exitedpercentage text
1	IsActive	5151	735	14.27%
2	NotActive	4849	1302	26.85%

- **Credit card ownership – HasCrCard vs NoCard shows very similar churn (about 20–21%), so impact is minimal by itself.**

	hascrcardstatus text	customercount bigint	exitedcount bigint	exitedpercentage text
1	HasCard	7055	1424	20.18%
2	NoCard	2945	613	20.81%

Composite Metrics

Engagement Score (ES)

Definition: The Engagement Score (ES) summarizes how actively a customer interacts with the bank based on their activity status, number of products, and credit card ownership.

- **Components and conditions:**

  IsActiveMember: Active customers are given 2 points, while inactive customers receive 0 points, reflecting that activity is a strong driver of lower churn.

  NumOfProducts: Customers with 2 products receive 2 points (most stable segment), those with 3–4 products receive 1 point (high usage but high churn), and customers with only 1 product receive 0 points.

  HasCrCard: Customers who hold a credit card receive 1 point, and those without a card receive 0 points, capturing an additional dimension of engagement.

Composite Metrics

Engagement Score (ES)

- **Score and buckets: The individual points are summed to form ES, which is then grouped into three buckets: 0–1 = Low Engagement, 2–3 = Medium Engagement, and 4–5 = High Engagement.**
- **Interpretation: Higher ES values indicate more engaged customers who typically show lower churn, while low ES customers combine weak product usage and low activity and therefore represent priority segments for engagement campaigns.**

	es_bucket text	totalcustomers bigint	exitedcustomers bigint	exitedpercentage text
1	Low	2571	970	37.73%
2	Medium	4886	854	17.48%
3	High	2543	213	8.38%

Composite Metrics

Risk Score (RS)

- **Definition: The Risk Score (RS) measures churn risk by combining the strongest individual drivers identified in the analysis: geography, age, activity status, and product portfolio.**
- **Components and conditions:**

  Geography: Customers in Germany receive 2 points due to significantly higher churn, whereas customers in France and Spain receive 0 points.

  Age group: Customers aged 36–55 receive 1 point, those aged 56–65 receive 2 points, and customers aged 18–35 or 65+ receive 0 or 1 point, reflecting the elevated churn observed in middle and pre-retirement age groups.

  Activity status: Inactive customers (IsActiveMember = False) receive 2 points, while active customers receive 0 points because inactivity is a strong churn signal.

  Number of products: Customers with 1 product receive 1 point, those with 2 products receive 0 points (most stable), and those with 3–4 products receive 2 points due to their extreme churn rates.

Composite Metrics

Risk Score (RS)

- **Score and buckets: The points are summed to obtain RS, which is grouped as 0–2 = Low Risk, 3–4 = Medium Risk, and 5+ = High Risk.**
- **Interpretation: Higher RS values indicate customers who combine multiple risk factors and therefore contribute disproportionately to total churn, making the High Risk bucket the primary target for retention actions.**

	rs_bucket text	totalcustomers bigint	exitedcustomers bigint	exitedpercentage text
1	Low Risk	4741	368	7.76%
2	Medium Ri...	3952	955	24.16%
3	High Risk	1307	714	54.63%

Composite Metrics

- **Financial Value Score (FVS)**
- **Definition: The Financial Value Score (FVS) captures the economic value of each customer by combining balance, estimated salary, and product depth into a single metric.**
- **Components and conditions:**

  Balance: Zero-balance customers receive 0 points; those with low balances (0–75k) receive 1 point; medium balances (75k–150k) receive 2 points; and high balances (above 150k) receive 3 points, reflecting increasing value held with the bank.

  Estimated salary: Customers in the low income 20th percentile receive 1 point, 20–40th percentile receive 2 points, 40–60th percentile receive 3 points, 60–80th percentile receive 4 points, and top 20% income customers receive 5 points, aligning higher earning capacity with higher potential value.

  Number of products: Customers with 1 product receive 1 point, those with 2 products receive 2 points, and customers with 3–4 products receive 3 points, since holding more products generally indicates higher revenue contribution.

Composite Metrics

- **Score and buckets: The three components are summed to form FVS, then grouped into 0–4 = Low Value, 5–7 = Medium Value, and 8+ = High Value.**
- **Interpretation: High FVS customers are financially significant for the bank; when intersected with the Risk Score, they identify "High Value–High Risk" customers who should be prioritized for retention strategies and personalized interventions.**

	fvs_bucket text	totalcustomers bigint	exitedcustomers bigint	exitedpercentage text
1	Low Value	2083	400	19.20%
2	Medium Val...	4947	887	17.93%
3	High Value	2970	750	25.25%

Actionable Recommendations

Product and Relationship Strategy

- **Promote 2-product bundles as the “sweet spot,” because customers with 2 products have the lowest churn (7.58%) and form a large base, so structured cross-sell from 1 to 2 products can significantly reduce churn.**
- **Investigate why customers with 3–4 products show extremely high churn (82.71–100%) through surveys, complaints analysis, and journey mapping, then simplify product combinations, pricing, and communication for these heavy users.**
- **Design retention offers that reward long-term, multi-year customers but avoid pushing them into overly complex product stacks, focusing instead on product quality and fit rather than quantity.**

Actionable Recommendations

Segment-Focused Retention Programs

- **Prioritize Germany for retention campaigns because its churn rate (32.44%) is about double that of France and Spain; use localized offers, language, and communication tuned to German customer expectations.**
- **Build targeted programs for middle-aged and pre-retirement customers (36–65), who show much higher churn, by offering financial planning, advisory services, and lifecycle-specific products that address their needs.**
- **Monitor and address higher female churn (25.07% vs 16.46% for males) by reviewing customer journeys, service touchpoints, and marketing messages to ensure they resonate and remove pain points for female customers.**

Actionable Recommendations

Engagement and Activity Management

- **Use activity status as a primary early-warning flag and set up triggers: when a customer becomes inactive (IsActiveMember = False), automatically enroll them in re-engagement campaigns such as reminders, relationship manager outreach, or tailored offers.**
- **Combine engagement and risk scores to identify “Low Engagement–High Risk” customers and prioritize proactive outreach (e.g., check-in calls, digital nudges, or fee waivers) before they decide to exit.**
- **Track key engagement metrics (logins, transactions, product usage) going forward and feed them into updated engagement scores for continuous monitoring rather than one-time analysis.**

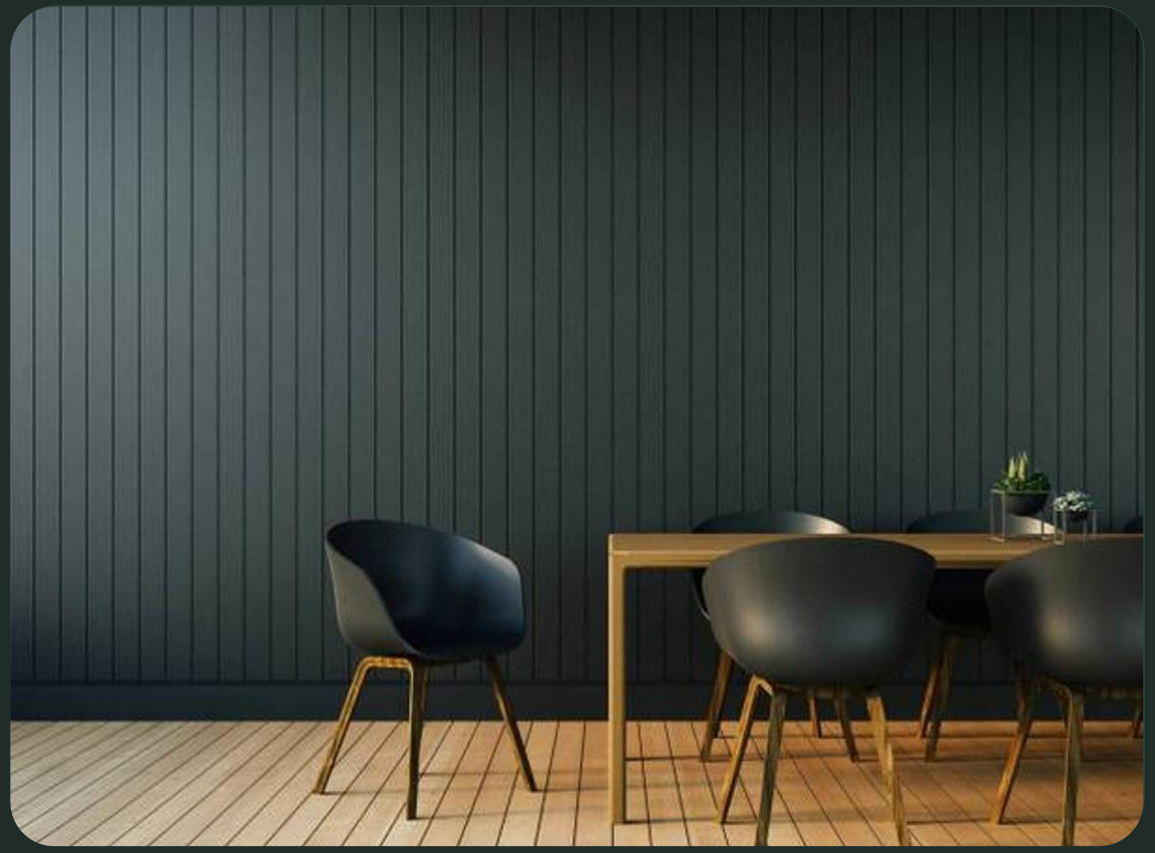
Actionable Recommendations

Value and Risk Prioritization

- **Use the Financial Value Score to identify High Value–High Risk customers (high balances/salaries, multiple products, but high RS) and assign them to premium retention actions such as dedicated relationship managers or personalized retention offers.**
- **For Low Value–High Risk customers, design scalable, low-cost digital interventions (emails, in-app messages) instead of expensive relationship management, ensuring efficient use of retention budgets.**
- **Periodically refresh RS and FVS (e.g., quarterly) and integrate them into dashboards so business teams can continuously track how many customers sit in each risk–value quadrant and how churn evolves after interventions.**

Summary

This churn study on 10,000 bank customers shows an overall churn rate of 20.37%, with risk heavily concentrated in specific segments rather than evenly spread across the base. Demographic factors such as geography (Germany at 32.44% churn) and age (36–65, especially 56–65 at 48.32%) significantly increase churn, while gender analysis reveals higher attrition among female customers. Financial variables like balance, salary, and credit score have weaker standalone impact, but behavioral indicators—particularly inactivity, complex product holdings (3–4 products), and low engagement—emerge as the strongest churn drivers, making composite scores (Engagement, Risk, and Financial Value) essential tools for prioritizing high-risk, high-value customers for targeted retention actions.



Thank you